

Student Performance Analysis Dashboard | Power BI Project Report

This report presents the complete development journey of the Student Performance Analysis Dashboard created using Power BI. The project focuses on transforming raw student performance data into interactive visual insights and meaningful educational analysis.

1. Project Objective

The main objective of this project was to analyze student academic performance using Power BI and identify how factors such as gender, ethnicity, parental education, and test preparation impact student scores in Math, Reading, and Writing.

2. Dataset Information

The dataset used in this project contains student-related educational information including:

- Student ID
- Gender
- Ethnicity
- Parental Education
- Test Preparation
- Math Score
- Reading Score
- Writing Score

3. Tools & Technologies Used

- Power BI
- Power Query Editor
- DAX (Data Analysis Expressions)
- Excel / CSV Dataset
- Data Visualization Techniques

4. Project Workflow

The project development process followed these steps:

1. Data Collection
2. Data Cleaning and Transformation
3. Creating Calculated Columns and Measures
4. Building Interactive Visualizations
5. Dashboard Design and Formatting
6. Data Analysis and Insights Generation

5. Data Cleaning & Transformation

The raw dataset was cleaned using Power Query Editor. Several calculated columns and transformations were created to improve analysis quality.

Created Columns:

- Total Mark
- Result Status
- Performance Category

Data types were corrected and unnecessary inconsistencies were removed.

6. DAX Measures Used

Important DAX calculations used in the project:

Pass Percentage:

Calculated the percentage of students who passed all subjects.

Average Total Marks:

Calculated overall average student performance.

Student Count:

Used for KPI cards and comparative analysis.

Performance Categories:

Students were categorized into Excellent, Good, Average, and Needs Improvement.

7. Dashboard Pages

Page 1 – Student Performance Overview

Page 2 – Subject-wise Performance Analysis

Page 3 – Gender-based Analysis

Page 4 – Parental Education Impact Analysis

Page 5 – Test Preparation Analysis

Page 6 – Key Insights & Conclusions

8. Visualizations Used

- KPI Cards
- Donut Charts
- Pie Charts
- Clustered Column Charts
- Stacked Column Charts
- Bar Charts
- Slicers and Filters

9. Problems Faced During the Project

While building the dashboard, several challenges were faced:

- Understanding DAX formulas and measures
- Creating correct Pass Percentage calculations
- Sorting performance categories properly
- Managing relationships between columns and measures
- Choosing suitable visualizations for analysis
- Formatting dashboard for better readability

10. Solutions Implemented

The following solutions were implemented:

- Used Power Query for data cleaning
- Created custom DAX measures for calculations
- Added slicers for interactive filtering
- Used conditional logic for Result Status and Performance categories
- Improved dashboard readability using proper themes and formatting
- Applied professional titles and KPI sections

11. Key Insights from the Dashboard

- Students who completed test preparation generally scored higher.
- Reading and Writing scores were stronger than Math scores overall.
- Parent education level positively influenced academic performance.
- Female students performed better in language-related subjects.
- Performance distribution highlighted different student learning levels.

12. Skills Improved Through This Project

- Data Cleaning
- Data Transformation
- DAX Calculations
- Dashboard Design
- Data Visualization
- Analytical Thinking
- Storytelling with Data

13. Future Improvements

Future improvements that can be added:

- Predictive Analysis
- Student Risk Prediction
- Attendance Analysis
- Drill-through Reports
- AI-based Educational Insights

14. Conclusion

This project helped strengthen practical Power BI skills and provided hands-on experience in data analysis, dashboard design, and business-style reporting. The Student Performance Dashboard demonstrates how educational data can be converted into actionable insights using interactive visualizations.

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